Tristan Hoellinger | Double Master's degree in Mathematical Modelling and Hybrid AI

108 avenue Jean-Jaurès – 92140 Clamart, France
0033 6 64 27 91 20 • tristan.hoellinger@insa-toulouse.fr • hoellin.github.io
in tristan-hoellinger

Professional experience

Master thesis **Châtillon**
ONERA - The French Aerospace Lab *February 2022 - Febryary 2023*
Intern in the NFLU (Digital methods for fluid mechanics) and MSAT (Advanced turbulence modeling and simulation) teams, in the DAAA departement. I develop new high-order numerical methods to better preserve vorticity in 3d turbulent flows simulation. Advisor : Jean-Baptiste Chapelier, jean-baptiste.chapelier@onera.fr

Assistant Bioinformatics Scientist **Toulouse**
Inserm, French National Institute for Health and Medical Research *September 2020 - January 2022*
I studied machine learning (XGBoost, deep neural networks) and heuristic methods to predict enhancer-gene relations in the human genome, based on high-throughput data. I presented my work at JOBIM conference 2021 (*Journées Ouvertes en Biologie, Informatique et Mathématiques* ; ranked 4th best poster among 46 for SFBI awards), and will soon publish my work as first author in a peer-reviewed journal. Advisor : Sarah Djebali, sarah.djebali@inserm.fr

Research Internship **Barcelone (Espagne)**
CRG - Centre for Genomic Regulation *July 2021 - August 2021*
International summer internship in Guigó's lab at CRG Barcelona. Developpement of a Nextflow pipeline based on Surrogate Variable Analysis (SVA) for the dentification and analysis of genetic variants associated with intron retention estimated from RNA-sequencing in blood cell types from the Blueprint Project. Advisor : Diego Garrido : diego.garrido@crg.eu

Research Internship **Toulouse**
IMT (Toulouse Mathematics Institute) *July 2020*
Stochastic calculus. I studied recent theoretical developments about the convergence of repeated quantum non-demolition measurements. Advisors : Tristan Benoist, tristan.benoist@math.univ-toulouse.fr and Clément Pellegrini, clement.pellegrini@math.univ-toulouse.fr

Blibliographic Internship **Toulouse**
IRAP - Institut de recherche en astrophysique et planétologie *April 2020 - June 2020*
French laboratory of space astrophysics operated by CNRS (French National Centre for Scientific Research) and Paul Sabatier University. I studied kinetic scale plasma turbulence in the solar wind. Supervisor : Philippe Louarn, philippe.louarn@irap.omp.eu

Research Internship **Strasbourg**
Inria, National Institute for Research in Digital Science and Technology *juin 2019 - août 2019*
AI trainee in MIMESIS team. I analysed the robustness of a deep learning method for data-driven biomechanical simulation, with respect to data sparsity and noises. As a result, we implemented an efficient transfer learning solution to learn new parameters with few data. The results were integrated in a peer-review paper. Advisor : Andréa Mendizabal, andrea.mendizabal@inria.fr

Farm Hand **Saint-Broing-les-Moines**
La Ferme aux 100 Blés *July 2017*
Agricultural operation. I used various tools (pneumatic hammer, freight elevators, riveting guns), ran a market stall and delivered products.

Formation

INSA Toulouse, ENSEEIHT **Toulouse**

Double MS in Mathematics, Modelling and Artificial Intelligence *2020-2022*

I pursue a double-curriculum entitled "Mathematical Modelling and Artificial Intelligence", which leads to a Diplôme d'ingénieur (Master degree) in Mathematical Modelling awarded by INSA Toulouse (Institut National des Sciences Appliquées de Toulouse), and a Diplôme d'ingénieur in Computer Science and Telecommunications awarded by ENSEEIHT. Graduation in March 2023. Main courses undertaken :

- optimization, stochastic optimization, data assimilation, optimal control
- signal and wavelets, advanced statistics, machine learning : clustering, kernel methods, CNN, recurrent networks, data assimilation networks, generative neural networks, reinforcement learning
- PDEs, finite elements and volumes, model coupling, high performance computing, resolution of large linear systems, parallel programming

Paris-Saclay University **Gif-sur-Yvette**

Gap year and academic semester *2019-2020*

I took a year off to study physics (hamiltonian mechanics, quantum physics, mathematical physics and cosmology, electrodynamics, astrophysics, statistical physics) for one semester at the Magistère de physique d'Orsay, after what I completed my gap year with 2 internships in physics.

INSA Toulouse **Toulouse**

Bachelor of Science *2017–2019*

First 3 years of a 5-year course leading to a Diplôme d'ingénieur. Specialisation in Applied Mathematics.

INSA Strasbourg **Strasbourg**

Propaedeutic year, Ranked top 10% *2016–2017*

Basic sciences.

Lycée Charles-Emile Freppel **Obernai**

High school diploma, Science, Highest Honors (18,1/20) *2013–2016*

Student projects

- Variational data assimilation to estimate river bathymetry from altimetric data.
- Kalman filters and ensemble Kalman filters : theory and application to linear and non-linear cases.
- Machine learning-based method to predict the energy of a molecule based on its geometric structure. Several approaches implemented, such as randomized Coulomb matrices and *Wavelet Scattering Transform*
- Implementation of a Variational Assimilation Network, with PyTorch, for supervised learning of Observed Dynamical Systems. Application to a simple linear dynamical system and to a Lorenz 40d system.

Languages

French: Mother Tongue

German: basics (A2)

English: Cambridge Linguaskill Business : CERCL C1+ (highest score)

Concrete Skills in Applied Mathematics

Modelling and numerical simulation: CFD, solving PDEs, ODE, discontinuous galerkine methods, lagrangian methods

Generalities: Optimal control, signal and image processing, smooth non-convex and convex non-smooth optimization, stochastic optimization

Statistics: inferential statistics, model selection, parametric tests, non-parametric tests, meta-modelling, gaussian processes, bayesian methods

Machine learning: clustering, kernel methods, bagging, boosting, neural networks (perceptrons, CNNs, recurrent networks, adversarial networks), data-assimilation networks, reinforcement learning

High Performance Computing: solving large sparse systems, parallel computing (OpenMP, MPI)

Concrete Skills in Computer Science

Programmation: C++, Fortran, Python (Keras, PyTorch, Tensorflow, pandas), Julia, R, bash, Awk, OCaml

Simulation: Python, Maple, Matlab

Version: Git, Github

Office: Suite Office, LaTeX, LibreOffice

Others: OpenMP, MPI, Docker, Kubernetes, AWS

Web: HTML5, CSS

Miscellaneous

- Former Mentor of 8 High School students for 6 month, in the context of the Equal Opportunity Program « Ô Talents », former President of the Theatre Troupe La Catin
- Hobbies : photography, sport (martial arts, running, hiking)

Publications

- As co-author : Data-driven simulation for augmented surgery, Andrea Mendizabal et al., 2020, Developments and Novel Approaches in Biomechanics and Metamaterials. Advanced Structured Materials, vol 132. Springer, Cham. `https://doi.org/10.1007/978-3-030-50464-9_5`
- As principal author : The genome-wide identification of enhancer/gene relationships : new reliable benchmark datasets needed to make progress in the field, T. Hoellinger, S. Djebali (published soon, `https://doi.org/10.1101/2022.10.12.511908`)

References

Internship Advisors

Sarah Djebali, permanent researcher at INSERM, sarah.djebali@inserm.fr

Jean-Baptiste Chapelier, permanent at ONERA, jean-baptiste.chapelier@onera.fr

Enseignants

Violaine Roussier-Michon, lecturer (Maîtresse de Conférence) at INSA Toulouse, roussier@insa-toulouse.fr